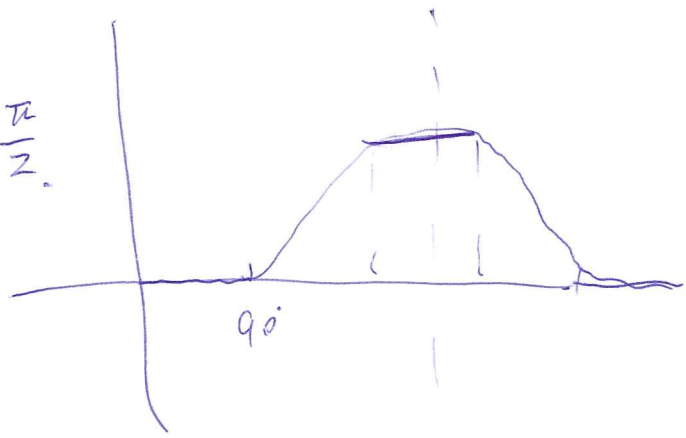


SCCA

Q1 Design Double dwell Cam.

rise 10mm 90°
 dwell 90°
 fall 10mm 90°
 dwell 90°

$h = 10\text{mm}$
 $\beta = 90^\circ = \frac{\pi}{2}$



1. min acceleration
 mod trap

b c d c v c a c j

Table 8-2

Rise

1 $y_1 =$
 2 $y_2 =$
 3 $y_3 =$
 4 $y_4 =$
 5 $y_5 =$

dwell

$y = 10$

fall

1 $y_1 = 10 - y_1$
 2 $y_2 = 10 - y_2$
 3
 4
 5

polynomial for Double-Dwell

Example 8-5

$$\beta = 90^\circ$$

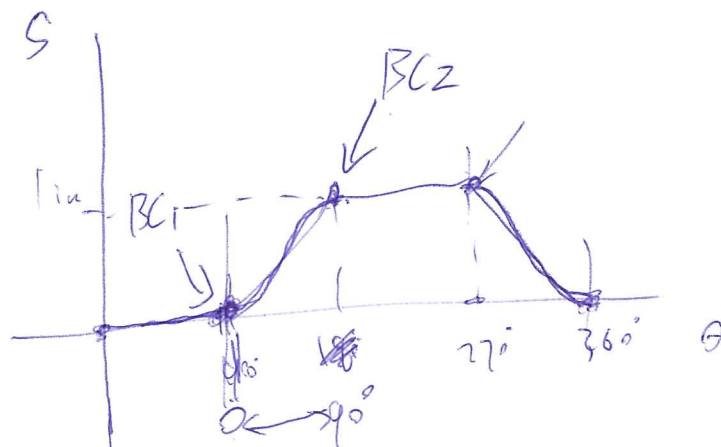
$$\omega = 2\pi \text{ rad/sec.}$$

dwell 90°

rise 1 in 90°

dwell 90°

fall 1 in 90°



rise when $\theta = 0^\circ$

$$\left. \begin{array}{l} S = 0 \\ V = 0 \\ a = 0 \end{array} \right\}$$

$\theta = \beta_1$

$$\left. \begin{array}{l} S = h \\ V = 0 \\ a = 0 \end{array} \right\}$$

fall when $\theta = 0^\circ$

$$\left. \begin{array}{l} S = h \\ V = 0 \\ a = 0 \end{array} \right\}$$

when $\theta = \beta_2$

$$\left. \begin{array}{l} S = 0 \\ V = 0 \\ a = 0 \end{array} \right\}$$

$$S = C_0 + C_1\left(\frac{\theta}{\beta}\right) + C_2\left(\frac{\theta}{\beta}\right)^2 + C_3\left(\frac{\theta}{\beta}\right)^3 + C_4\left(\frac{\theta}{\beta}\right)^4 + C_5\left(\frac{\theta}{\beta}\right)^5$$

$$V = \frac{1}{\beta} \left[C_1 + 2C_2\left(\frac{\theta}{\beta}\right) + 3C_3\left(\frac{\theta}{\beta}\right)^2 + 4C_4\left(\frac{\theta}{\beta}\right)^3 + 5C_5\left(\frac{\theta}{\beta}\right)^4 \right]$$

$$a = \frac{1}{\beta^2} \left[2C_2 + 6C_3\left(\frac{\theta}{\beta}\right) + 12C_4\left(\frac{\theta}{\beta}\right)^2 + 20C_5\left(\frac{\theta}{\beta}\right)^3 \right]$$

sub BC into 3 equations

$$\cancel{14C_3} \quad C_0 = 0 \quad C_1 = 0 \quad C_2 = 0$$

$$S \Rightarrow \cancel{C_3} \quad C_3 + C_4 + C_5 = h \quad \text{--- (1)}$$

$$V \Rightarrow \frac{1}{\beta} (3C_3 + 4C_4 + 5C_5) = 0 \quad \text{--- (2)}$$

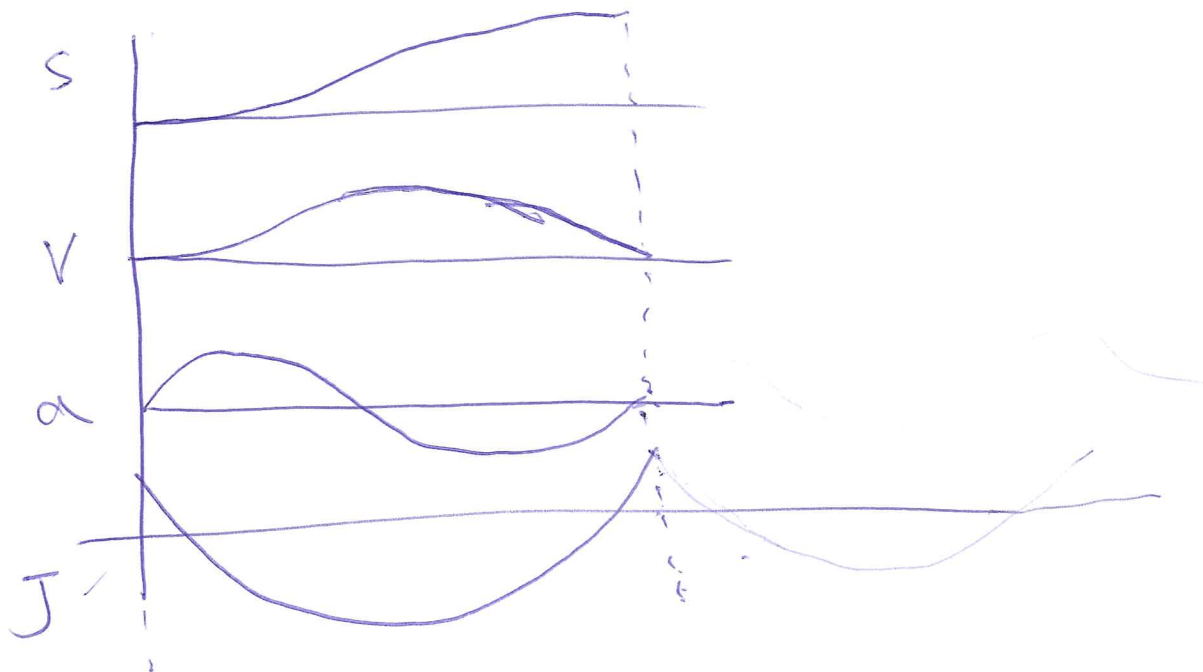
$$a \Rightarrow \frac{1}{\beta^2} (6C_3 + 12C_4 + 20C_5) = 0 \quad \text{--- (3)}$$

$$C_3 = 10h \quad C_4 = -15h \quad C_5 = 6h$$

$$S = h \left[10 \left(\frac{\theta}{\beta} \right)^3 - 15 \left(\frac{\theta}{\beta} \right)^4 + 6 \left(\frac{\theta}{\beta} \right)^5 \right]$$

$$V = \dots$$

$$a = \dots$$



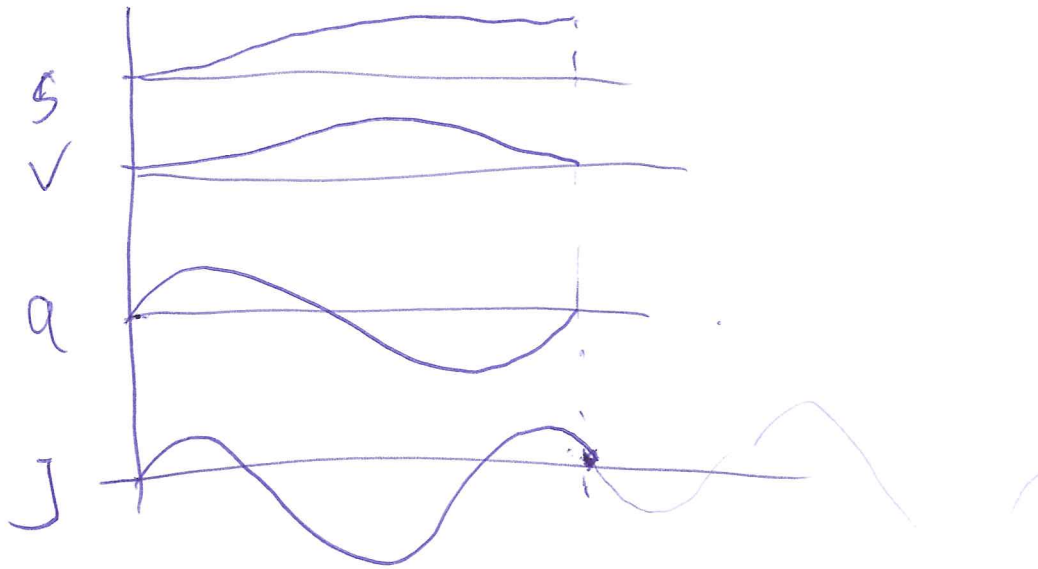
How do we mod the poly function to make jerk curve looks nice.

$$\theta=0 \quad s=0 \quad v=0 \quad a=0 \quad \dot{j}=0$$

$$\theta=\beta \quad s=h \quad v=0 \quad a=0 \quad j=0$$

8 BCs.

Use 4-5-6-7 polynomial function.



Single Dwell Cam Design.

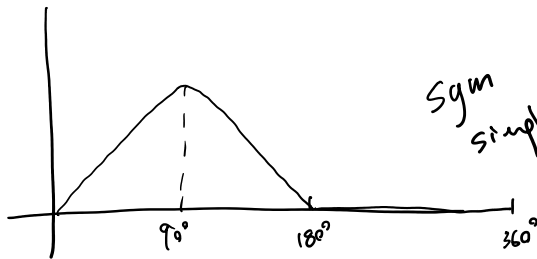
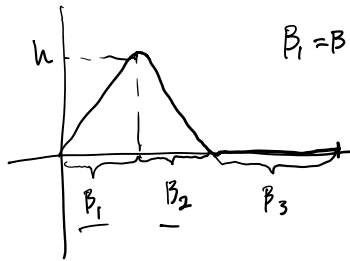
rise h, β_1

fall h, β_2

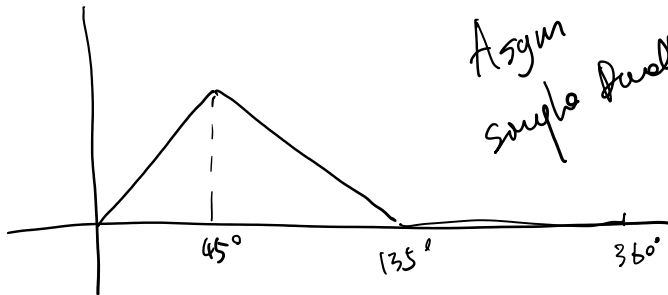
dwell β_3

$\beta_1 \neq \beta_2$ Asym

$\beta_1 = \beta_2$ sym



Sym
single Dwell



Asym
single Dwell

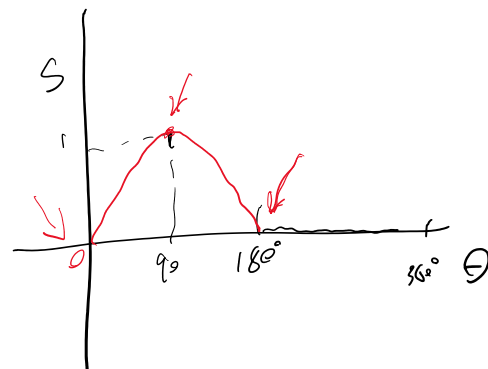
Designing a Polynomial for the Symmetrical rise-fall
single Dwell Case

Example 8-8

rise-fall $1, 90^\circ$ $1, 90^\circ$

dwell $0, 180^\circ$

Cam $\omega = 15 \text{ rad/sec}$



BCs

at start of the rise: $\theta = 0$ $s = 0$ $v = 0$ $a = 0$

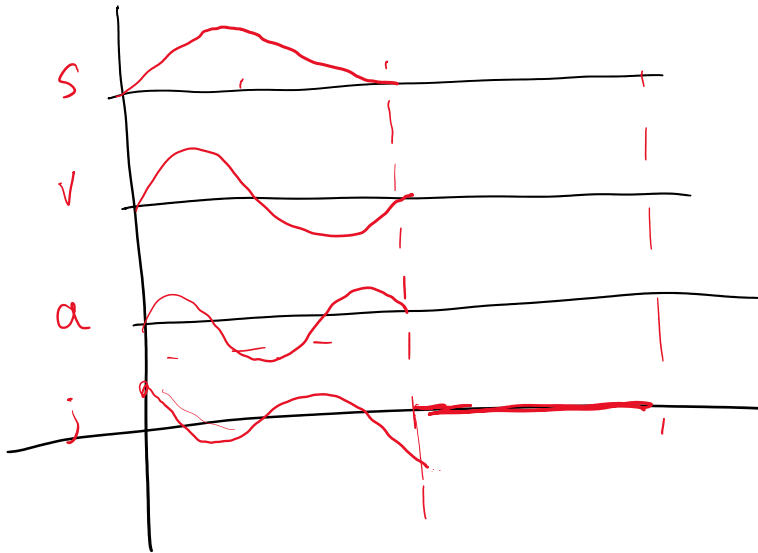
peak: $\theta = \frac{\pi}{2}$ $s = 1$

end of fall: $\theta = \pi$ $s = 0$ $v = 0$ $a = 0$

end of fall: $\theta = \pi$ $s=0$ $v=0$ $a=0$

$$\Rightarrow S = h \left[64 \left(\frac{\theta}{\pi} \right)^3 - 192 \left(\frac{\theta}{\pi} \right)^5 + 64 \left(\frac{\theta}{\pi} \right)^6 \right]$$

$\checkmark$
 a
 J



single dwell

Jerk - not
need to
return to
zero.

Asymmetry rise-fall single dwell $\rightarrow$ Polynomial.

Example 8-9

rise-fall : rise 1, 45° , fall 1, $135^\circ \Rightarrow 180^\circ$

dwell 0, 180°

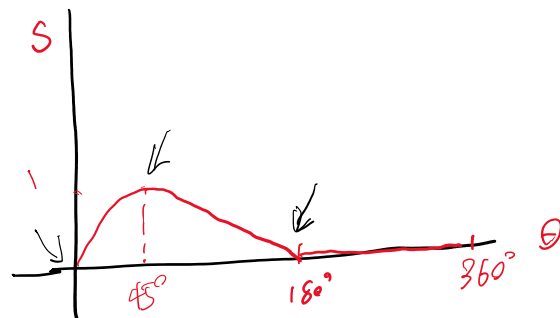
cam ω 15 rad/sec.

$$\theta = 0 \quad s = v = a = 0$$

$$\theta = \frac{\pi}{2} \quad s = v = a = 0$$

$$\theta = \frac{\pi}{4} \quad s = 1$$

Seven BCs



$$S = h \left[151.704 \left(\frac{\theta}{\pi} \right)^3 - 455.111 \left(\frac{\theta}{\pi} \right)^4 + 455.111 \left(\frac{\theta}{\pi} \right)^5 - 151.704 \left(\frac{\theta}{\pi} \right)^6 \right]$$

$$V = \dots$$

$V = \dots$
 $a = \dots$
 $J = \dots$

V at 45° not zero



seven

BCs

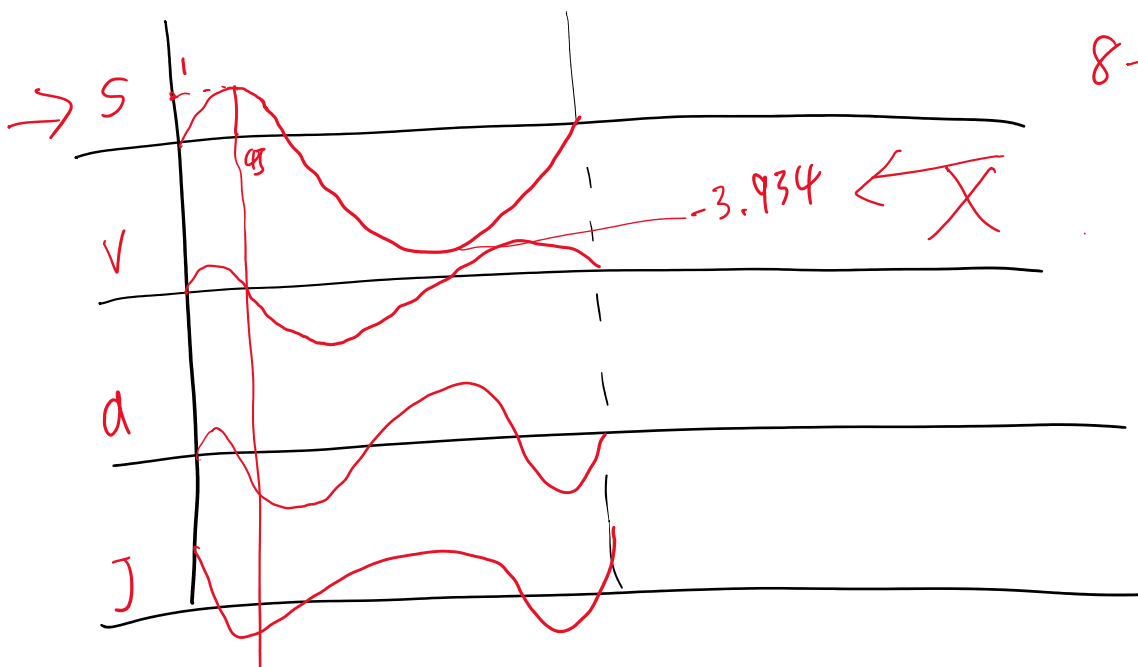
we forgot
to add

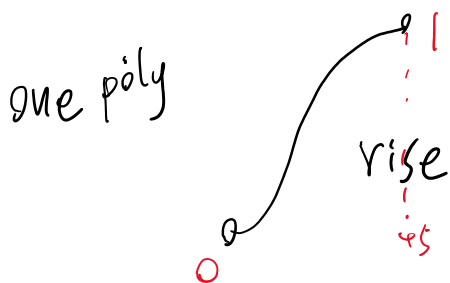
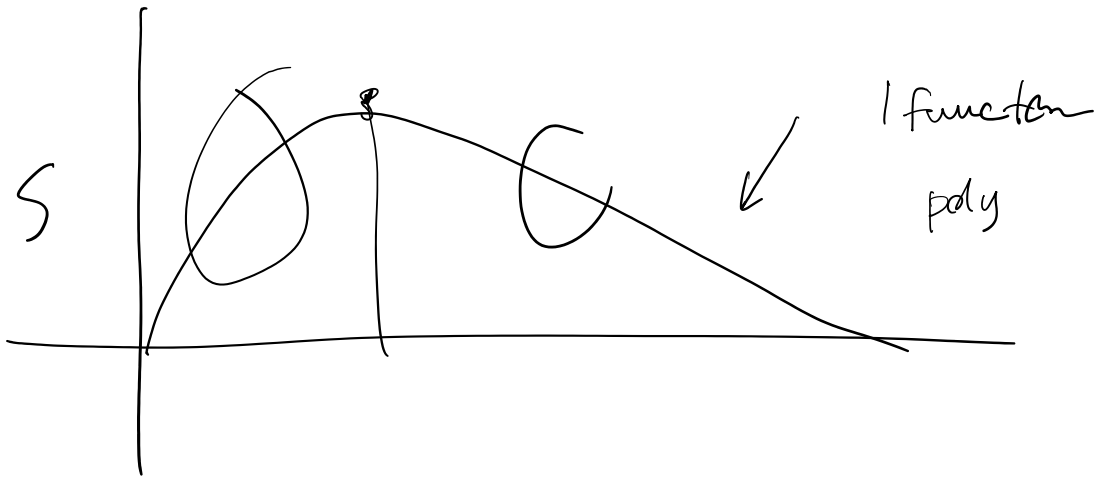
$V=0$ at 45°

add

$V=0$ at 45°

8 $\rightarrow$ BCs



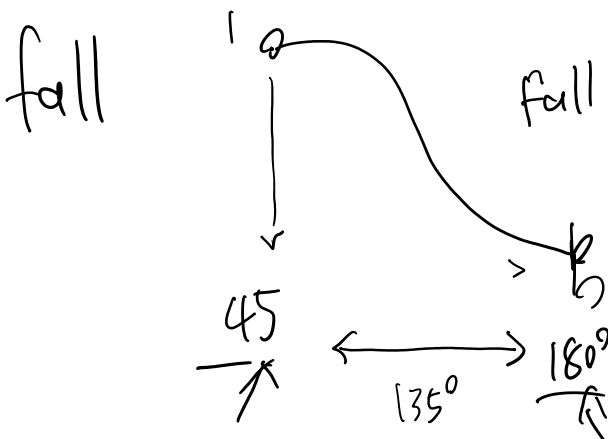


another poly



rise BC: $\theta = 0^\circ$ $S = 0$ $V = 0$ $\alpha = 0$
 $\theta = 45^\circ$ $S = 1$ $V = 0$ } 5 BCs

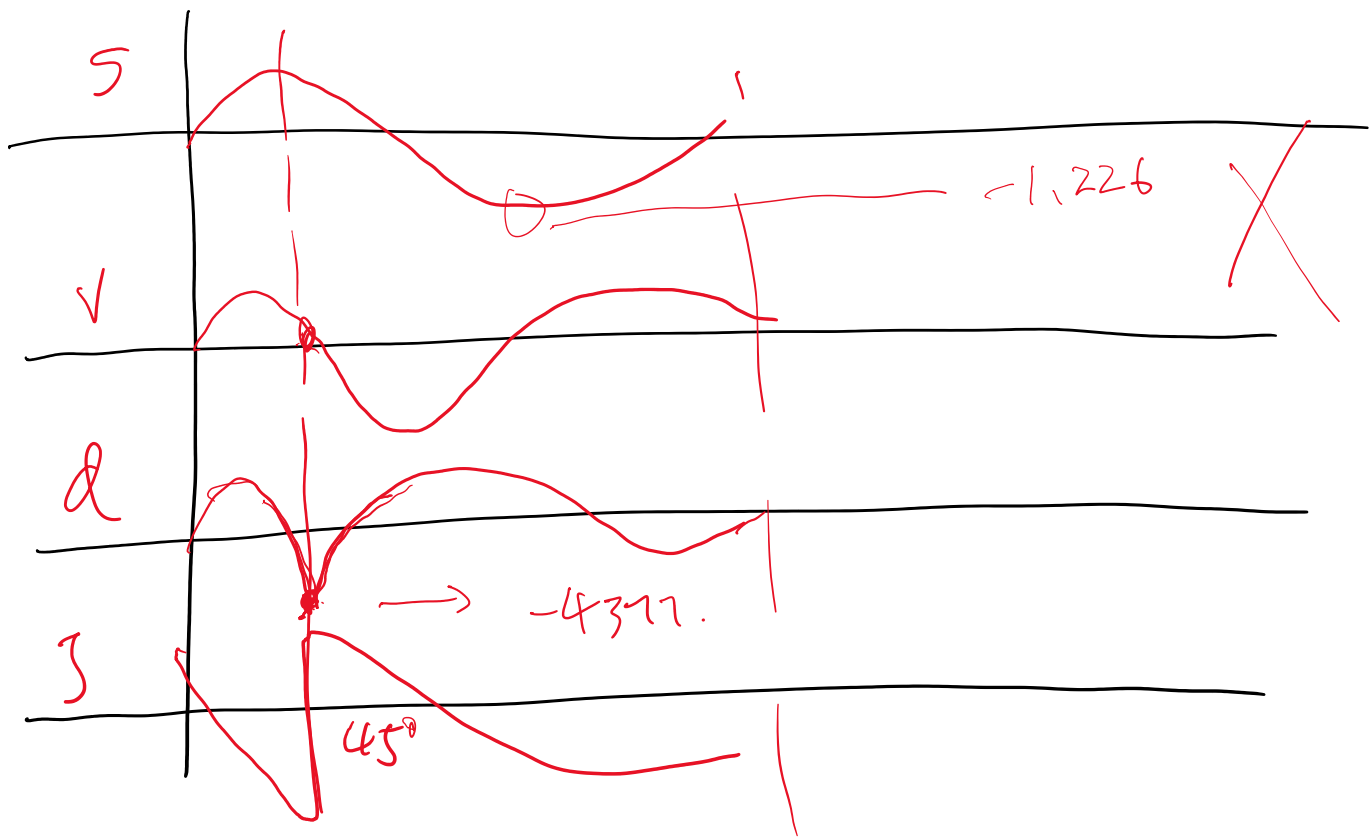
rise $S = h \left[4 \left(\frac{\theta}{B} \right)^3 - 3 \left(\frac{\theta}{B} \right)^4 \right]$ Find α @ 45°
 to connect two polys
 $\alpha = -4377.11$ @ $\theta = 45^\circ$



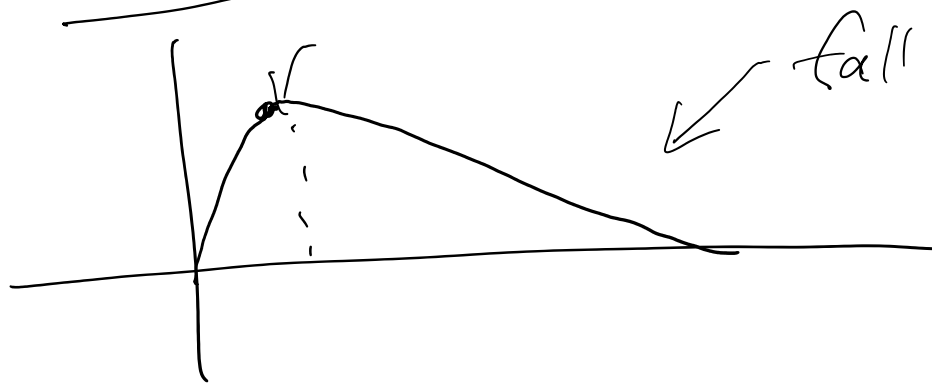
BCs $\theta = 45^\circ$ $S = 1$ $V = 0$
 $\alpha = -4377.11$

$\theta = 180^\circ$ $S = 0$ $V = 0$
 $\alpha = 0$

fall: $S = h \left(1 - 54 \left(\frac{\theta}{B} \right)^2 + 152 \left(\frac{\theta}{B} \right)^3 - 147 \left(\frac{\theta}{B} \right)^4 + 48 \left(\frac{\theta}{B} \right)^5 \right)$



use rise function to calculate the accele-



high a .

Calculate fall first

Use BC $\theta = 45^\circ$ $s = 1$ $v = 0$
 $\theta = 180^\circ$ $s = 0$ $v = 0$ $a = 0$ } 5 BCs

Find a @ 45°

r , r^2 , r^3 , r^4] , $\theta = 45^\circ$

Fall $S = h \left[1 - 6 \left(\frac{\theta}{\beta} \right)^2 + 8 \left(\frac{\theta}{\beta} \right)^3 - 3 \left(\frac{\theta}{\beta} \right)^4 \right]$ $\theta = 45^\circ$
 $S' = a$

$a @ 45^\circ = -486.4 \text{ in/sec}^2$

Rise BCs $\theta = 0 \quad S = 0 \quad V = 0 \quad a = 0$

6 BCs $\theta = 45^\circ \quad S = h \quad V = 0 \quad a = -486.4$

Rise $S = h \left[9.333 \left(\frac{\theta}{\beta} \right)^3 - 13.667 \left(\frac{\theta}{\beta} \right)^4 + 5.333 \left(\frac{\theta}{\beta} \right)^5 \right]$.

